FREE GUIDE · VERSION 1.0

OCTOBER 2026

PAVAN KROVVIDI FINGERWAVE NEURAL NETWORK

The *FingerWave* Guide



Every life has a unique print.
So does every prediction.

A neural network inspired by how fingerprints form in the womb: the theory, the mathematics and how to use it in seven industries.

- From biology to AI
- 10 equations, explained simply
- Benchmark evidence
- 7 industry playbooks

Open source · `pip install pavan-krovvidi-fingerwave`



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START HERE

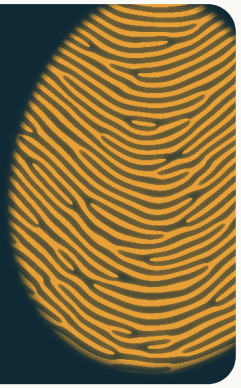
About this guide

This guide explains the **Pavan Krovvidi FingerWave Neural Network** from first principles: where the idea comes from, the concepts and mathematics behind it, how it is built, what it achieved, and how it can be applied across industries. It is written for three kinds of readers, and each can take a different path through it.

IF YOU ARE…	START WITH	THEN READ
A business or product leader	Chapter 1 (one-page summary)	Part B: the fit test and your industry chapter, then Chapter 17
An engineer or data scientist	Chapters 1, 3 and 5	Chapters 4, 7 and 8, then the industry chapter closest to your data
A researcher or student	Chapters 2 to 4	Chapters 5 to 7, the references and the interview FAQ

THE ONE-SENTENCE VERSION

FingerWave is a neural network that repairs damaged structural data the way a fingertip grows its ridges: it starts from the parts it can trust, lets information travel along the structure like a wave, and refuses to invent or lose the special points that define the pattern.



Everything in Part A is backed by working, published code: the package is on PyPI (`pip install pavan-krovvidi-fingerwave`), the source is on GitHub, and the research report has a permanent DOI (10.5281/zenodo.23766123). All numbers quoted in this guide come from that report and from runs of the released package.

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CHAPTER 01 · PART A · THEORY

1

FingerWave in one page

Thirty seconds to understand everything that follows.



31%
fewer topological errors than a same-size U-Net

96%
fingerprint class accuracy from recovered structure

+29
extra parameters over the U-Net (118,463 vs 118,434)

3 / 3
random seeds where FingerWave beat every U-Net run

The problem

Most AI models treat an image as a grid of independent pixels. When part of the input is noisy or missing, they fill the gap with whatever looks plausible locally. For many kinds of data this produces **structural hallucinations**: patterns that look fine pixel by pixel but are impossible as a whole. In a fingerprint, that means extra cores and deltas that do not exist. In a brain scan, it can mean nerve fibres that turn in impossible ways. In a flow field, it means vortices that appear from nowhere.

The idea

Nature already solves this problem every time a baby's fingerprints form. Ridges start at a few reliable places on the fingertip and travel inward as waves. Where waves meet, they settle into a small, fixed number of special points. FingerWave copies that process in three steps:

1. **Seed** BOUNDARY SEEDING
   Start from what you can trust. A learned confidence map anchors the reliable pixels first.

2. **Wave** WAVEFRONT PDE LAYER
   Spread information along the structure, from trusted regions into damaged ones.

3. **Conserve** TOPOLOGICAL LOSS
   Never invent or lose a singular point. A differentiable rule enforces this during training.

> *The biggest gain didn't come from a bigger model. It came from deciding where to start.*

The evidence

On a benchmark of 500 degraded fingerprints with exact ground truth, FingerWave reduced the Topological Error Rate (the share of prints with the wrong number of cores or deltas) from **5.8% to 4.0%** compared with a U-Net of the same size, gave the lowest orientation error inside hidden regions, and was about five times more consistent across random seeds. Removing the seeding step made errors jump to 37.7%, which shows that the biological idea of starting from trusted regions is what makes the method work.

Where it applies

FingerWave is built for data that **form a map**, **carry a direction or structure**, and are **partly missing or noisy**. That description fits fingerprints and forensics, medical imaging (brain fibres, retinal vessels, heart muscle), drug discovery (electrostatic fields, cryo-EM maps), automotive engineering (airflow, battery heat), finance (volatility surfaces, risk maps), retail and logistics (customer and robot flow), and energy and earth science (seismic layers, satellite gaps). Part B explains each of these in detail.

PART A

A Theory



From a fingertip in the womb to a neural network: the biology, the concepts, the mathematics, the architecture and the evidence.

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CHAPTER 02 · PART A · THEORY

2

How nature draws a fingerprint

Weeks before birth, waves draw a pattern that lasts a lifetime.

Fingerprints are laid down between roughly the **10th and 17th weeks of pregnancy**, on small temporary swellings of the fingertip called **volar pads**. In 2023, Glover and colleagues showed in the journal *Cell* that the ridges are produced by a **Turing-type reaction–diffusion system**. Two signalling pathways, WNT and EDAR, act as *activators* that promote ridge formation, while BMP signalling acts as an *inhibitor* that suppresses it nearby. The interplay of a short-range activator and a longer-range inhibitor is exactly the mechanism Alan Turing proposed in 1952 to explain stripes and spots in nature.

The crucial detail for FingerWave is **where** and **when** ridges begin. They do not appear everywhere at once. Ridge formation starts at a small number of **initiation sites**: the apex of the fingertip, the area near the nail, and the crease of the last finger joint. From there, ridges spread across the pad as **travelling waves**.

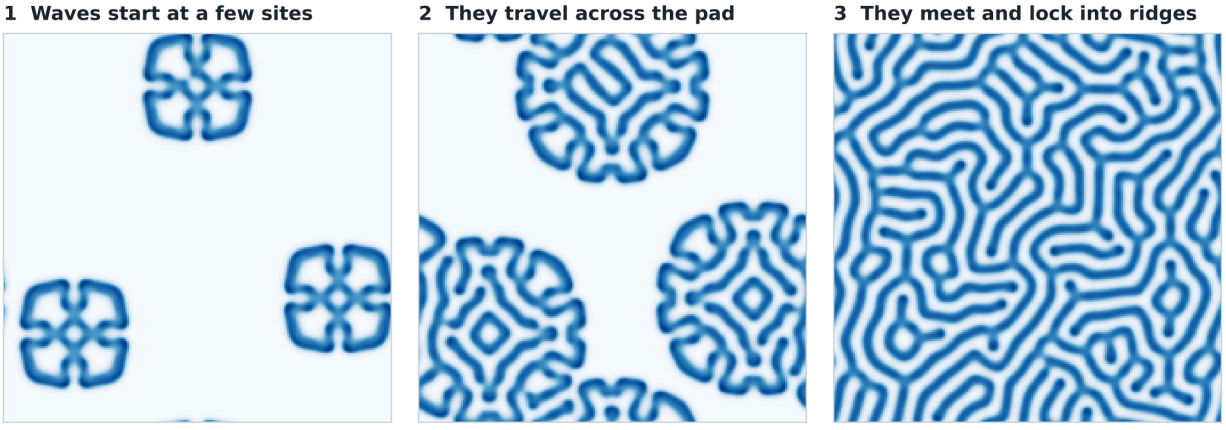


FIGURE 1 · A Turing reaction–diffusion simulation (Gray–Scott model) started from three sites. Patterns spread outward as waves, meet, and lock into a labyrinth of ridges. This is an illustration of the mechanism, not a model of a real finger.

Where the waves meet

When two wavefronts with different ridge directions collide, the field has to reconcile them. The compromise creates the **singular points** used by every fingerprint expert:

- **Core**: a place where ridges turn back on themselves, like the top of a loop. Its topological index is **+½**.

- **Delta**: a place where three ridge systems meet in a small triangle. Its index is **−½**.

The size and shape of the volar pad and the timing of each wave decide whether the final pattern becomes an **arch** (no cores or deltas), a **loop** (one core and one delta) or a **whorl** (two cores and two deltas). Because these are decided by small differences in timing and geometry, even identical twins have different fingerprints.

> ***Ridges don't appear everywhere at once. They start at a few reliable places and travel inward as waves.***

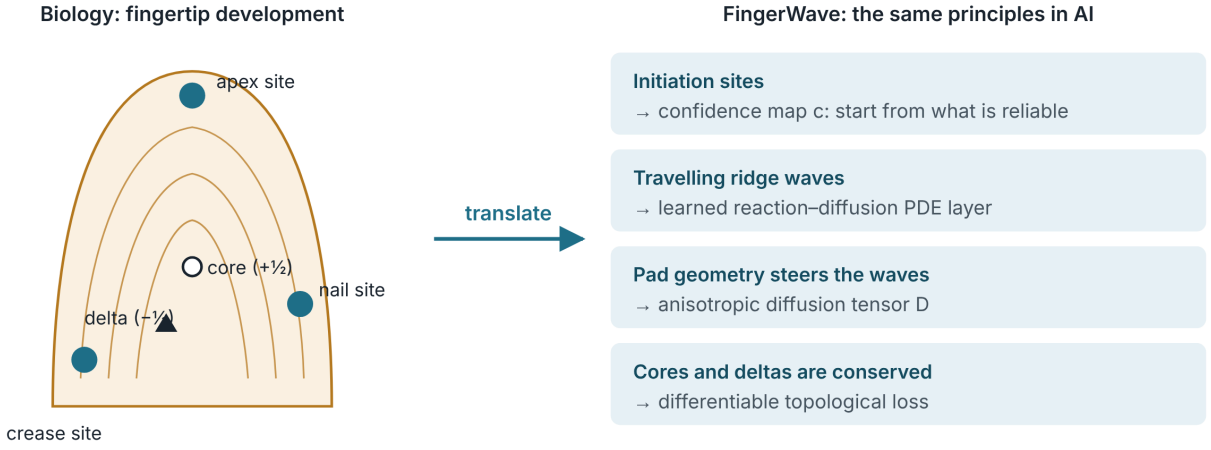


FIGURE 2 · The developmental principles behind fingerprints (left) and the FingerWave component that each one inspired (right).

KEY IDEA

The design principle

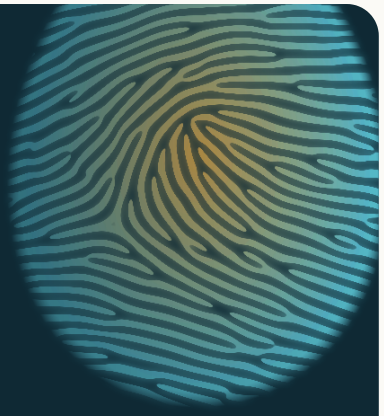
An AI model that fills in missing structure should behave like a developing fingertip: start from the places it trusts, let information travel along the structure instead of across it, and respect the rules that govern how singular points are created and destroyed.

CHAPTER 03 · PART A · THEORY

3

Seven ideas you need

Fields, orientations, defects and waves, with everyday examples.



FingerWave combines ideas from biology, physics and deep learning. None of them is difficult on its own. This chapter introduces each one with an everyday example before the mathematics in Chapter 4.

3.1 A field

A **field** is a value attached to every point of a space. A weather map that shows the temperature at every location is a field. So is a map of wind directions, the brightness of every pixel in a photo, or the direction of the ridges at every point of a fingerprint. FingerWave takes a damaged field as input and returns a repaired field as output.

3.2 Orientation is not direction

A wind arrow has a head and a tail: wind blowing north is different from wind blowing south. A fingerprint ridge has no head or tail. A ridge at 30° is exactly the same ridge as one at 210°. Mathematicians call this kind of headless direction an **orientation**, and a field of orientations a **director field** (the word comes from liquid-crystal physics).

EXAMPLE

Everyday example

A two-way road has an orientation (it runs north–south), but a one-way street has a direction (it runs north). Fingerprint ridges, nerve fibres, wood grain and crop rows are two-way roads. Wind, water currents and customer movement are one-way streets.

3.3 The doubled-angle trick

Computers struggle with headless angles, because 30° and 210° look very different as numbers. The fix is to **double the angle**: 2 × 30° = 60° and 2 × 210° = 420° = 60°. After doubling, both descriptions of the same ridge land on the same point of a circle. FingerWave stores this doubled angle as a complex number Ψ, which makes the field smooth and easy to process.

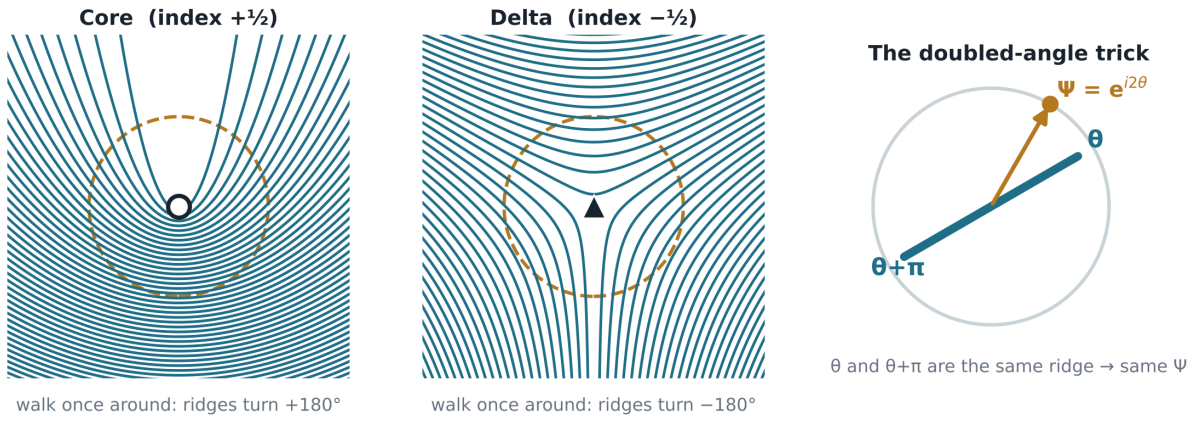


FIGURE 3 · Left and centre: ridge patterns around a core and a delta. Walking once around the dashed circle, the ridges turn by +180° or −180°. Right: doubling the angle maps θ and θ+π to the same point Ψ.

3.4 Topological defects

Walk in a small circle around any point of a fingerprint and keep track of how the ridge direction turns. Usually it turns a bit and comes back: the total is 0°. Around a core, it turns by exactly +180°; around a delta, by −180°. Dividing by 360° gives the **index**: +½ for a core and −½ for a delta. Points with a non-zero index are called **topological defects**.

The key property is that the index is a **whole-number-like quantity**: it cannot change a little bit. A small amount of noise cannot turn a core into half a core. Defects can only appear or disappear in matching pairs (a core with a delta). This is why fingerprint topology is so robust, and why an AI model that creates a spurious pair is making a **qualitative** mistake, not a small numerical one.

EXAMPLE

Everyday example

The crown of hair on a person's head is a topological defect of the hair-direction field. You can comb the hair, but you cannot remove the crown without creating another one somewhere else.

3.5 Reaction–diffusion

Diffusion spreads things out: a drop of ink in water slowly blurs. **Reaction** creates or destroys things locally: a chemical reaction makes or uses up a substance. When the two act together, they can produce stable patterns such as stripes, spots and waves. This is how leopards get their spots, how zebrafish get their stripes, and how fingertips get their ridges. FingerWave's core layer is a learnable reaction–diffusion system.

3.6 Confidence

Not all input pixels deserve the same trust. In a fingerprint, a clean region with sharp ridges is reliable, while a smudge or a missing band is not. FingerWave learns a **confidence map** c, a number between 0 and 1 for every pixel. High-confidence pixels act like the initiation sites of a real fingertip: they hold their value and send information outward. Low-confidence pixels are filled in by the waves.

EXAMPLE

Everyday example

A radiologist reading a blurry scan does not start in the blurriest region. They anchor on the clearest anatomy first and reason outward from there. FingerWave does the same thing automatically.

3.7 A PDE inside a neural network

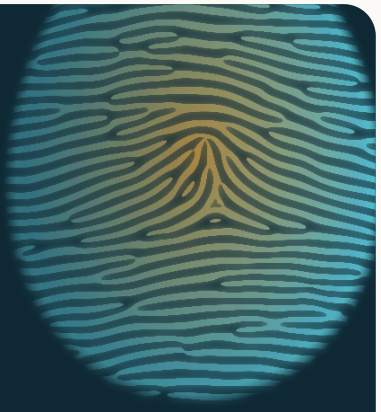
A **partial differential equation (PDE)** describes how a field changes over time based on its values and slopes nearby. Heat flow, sound and water waves are all described by PDEs. A computer simulates a PDE by repeating a small update step many times. FingerWave puts such a simulation **inside** the network as a layer. Each step is like one tick of a clock, and the coefficients that control the speed and shape of the waves are **learned from data** instead of being fixed by hand.

GOOD TO KNOW

Why this matters

A PDE layer carries a built-in prior about how information should spread. A plain convolutional network has to learn this from examples, and it can learn shortcuts that break the structure. A PDE layer cannot spread information in ways the equation does not allow.

CHAPTER 04 · PART A · THEORY



The mathematics, one formula at a time

Ten equations, each with its symbols and a plain-language meaning.

This chapter presents every equation in FingerWave. Each one comes with a table of symbols and a plain-language explanation. The notation follows the research report and the released code, so you can match each formula to a line of Python.

4.1 The field representation

$$\Psi = A \cdot e^{i2\theta} = u + i\,v, \text{ with } u = A\cos 2\theta,\ v = A\sin 2\theta \qquad (1)$$

θ	Ridge orientation at a pixel, in radians (θ and θ+π are the same ridge)
A	Amplitude, between 0 and 1: how clear the ridge structure is
u, v	The two real channels that the network actually stores
Ψ	The complex doubled-angle field (Greek letter psi)

In plain words: each pixel stores a small arrow of length A pointing at twice the ridge angle. Doubling removes the head-or-tail ambiguity (Section 3.3). As a bonus, a fingerprint core (+½) becomes an ordinary +1 vortex of Ψ and a delta (−½) becomes a −1 vortex, which are much easier to handle mathematically. To read the ridge angle back, use θ = ½ · atan2(v, u).

4.2 The topological index

$$k = (1 / 2\pi) \oint d\theta \text{ (ridge index, } \pm\tfrac{1}{2} \text{ for cores and deltas)} \quad (2)$$

$\oint$	Integral once around a small closed loop
$d\theta$	The change of ridge angle along the loop
k	Index of the point inside the loop

In plain words: count how many full turns the ridge direction makes as you walk around a point. On a pixel grid, FingerWave does this on every 2 × 2 block of pixels: it adds up the four wrapped angle changes around the block and divides by 2π. This exact counter is used to **evaluate** models. It produces whole numbers, so it has no useful gradient and cannot be used for training directly. Section 4.6 solves that problem.

4.3 The diffusion tensor

$$D = d_{\perp} I + (d_{\parallel} - d_{\perp}) \cdot c \cdot t\, t^{T},\ t = (\cos\theta, \sin\theta) \quad (3)$$

$d_{\parallel}$	Diffusion speed along the ridges (learned)
$d_{\perp}$	Diffusion speed across the ridges (learned)
I	The 2 × 2 identity matrix
t	Unit vector along the ridge
c	Local coherence: how clearly a ridge direction exists

In plain words: D is a 2 × 2 matrix that says how fast information may travel in each direction. Along the ridge it travels at speed $d_{\parallel}$; across the ridge, at $d_{\perp}$. If $d_{\parallel} > d_{\perp}$, information prefers to flow along ridges, the way a real ridge wave follows the shape of the pad. Where there is no clear ridge ($c \approx 0$), D becomes the same in all directions.

GOOD TO KNOW

An implementation detail that matters

Computing θ with an arctangent fails where Ψ = 0 (exactly at a defect), because the gradient becomes infinite. FingerWave avoids the arctangent entirely with the identity $t\,t^{T} = \tfrac{1}{2}(I + [[\cos 2\theta, \sin 2\theta], [\sin 2\theta, -\cos 2\theta]])$, and cos 2θ and sin 2θ are just u/A and v/A. So D is computed directly from Ψ, and gradients stay finite everywhere.

4.4 The wavefront PDE

$$\partial\Psi/\partial t = \nabla\cdot(D\,\nabla\Psi) + (\alpha + i\beta)\,\Psi - (1 + i\gamma)\,|\Psi|^{2}\,\Psi \qquad (4)$$

$\partial\Psi/\partial t$	How the field changes in one tick of time
$\nabla\cdot(D\,\nabla\Psi)$	Anisotropic diffusion: spreading along the structure
α	Growth rate: lets weak structure strengthen
β, γ	Rotation terms that create travelling and spiral waves
$\lvert\Psi\rvert^{2}\Psi$	Saturation: stops the amplitude from growing forever

In plain words: this is the **complex Ginzburg–Landau equation** with anisotropic diffusion, the standard physics model for travelling waves and spiral patterns. The first term spreads the field along the ridges. The second term makes weak regions grow. The third term caps the growth so the amplitude settles near √α. Together they turn a noisy, partly empty field into a clean, wave-like pattern, the same way reaction–diffusion turns a smooth fingertip into ridges.

The diffusion term is computed in **conservative flux form**: the code computes how much flows across each face between two pixels, and each pixel gains exactly what its neighbour loses. This guarantees that the layer only moves information around and never creates it out of nothing. In the verification tests, the total field changed by only 1.2×10^{-8} (relative) over 500 steps.

4.5 Boundary seeding

$$\partial\Psi/\partial t = P(\Psi) + \lambda \cdot c \cdot (s - \Psi) \qquad (5)$$

$P(\Psi)$	The wavefront dynamics of Equation 4
s	Seed field predicted by the encoder
c	Confidence map, between 0 and 1
λ	Seeding strength (learned, bounded so that $\lambda\cdot\Delta t \le 0.8$)

In plain words: at every time step, each pixel is gently pulled towards the encoder's guess s, in proportion to how much the network trusts that pixel. Where c ≈ 1, the pixel behaves like an initiation site and holds its value. Where c ≈ 0, the pull disappears and the waves fill the pixel in from its trusted neighbours. This one term is the most important part of FingerWave: removing it raised the error rate from 4.0% to 37.7%.

KEY IDEA

Why a gentle pull and not a hard overwrite

An early version simply replaced trusted pixels with the seed at every step. That blocked learning almost completely: the PDE coefficients received gradients of about 10^{-25}. Turning the overwrite into a forcing term (Equation 5) let the gradients flow, and training worked normally.

> ***Remove the seeding term, and the error rate jumps from 4.0% to 37.7%.***

4.6 One time step and its stability

$$\Psi^{*} = \Psi^{n} + \Delta t\,[\ \nabla\cdot(D\,\nabla\Psi^{n}) + (\alpha + i\beta)\,\Psi^{n}\ ] \tag{6a}$$

$$\Psi^{n+1} = e^{-i\gamma\Delta t|\Psi^{\wedge}\{n}|^{2}\}\cdot\Psi^{*} / (1 + \Delta t\,|\Psi^{n}|^{2}) \tag{6b}$$

$$\Psi^{n+1} \leftarrow \Psi^{n+1} + \Delta t\cdot\lambda\cdot c\cdot(s - \Psi^{n+1}) \tag{6c}$$

Δt	Time step, set to 0.2 / d_{max}
d_{max}	Upper bound on the diffusion speeds (1.0 by default)
n	Step counter

In plain words: each step first applies diffusion and growth (6a), then applies the saturation and rotation in a **semi-implicit** way (6b), then applies the seeding pull (6c). The division in 6b is the trick that keeps the cubic term from exploding: the bigger the amplitude, the stronger the damping. An explicit diffusion step on a unit grid is stable when $\Delta t \cdot d \le ¼$. FingerWave bounds $d_{\parallel}$ and $d_{\perp}$ below d_{max} with a sigmoid and uses $\Delta t = 0.2 / d_{max}$, so the layer stays inside that limit for any learned value. In the tests, an unprotected scheme reached 10^{40} and then failed, while the FingerWave layer stayed near 1.

4.7 The differentiable topological loss

$$E_{GL} = \langle|\nabla\Psi|^{2}\rangle + \langle(1 - |\Psi|^{2})^{2}\rangle / (2\varepsilon^{2}) \tag{7}$$

$$\rho = (\partial_{x}u\,\partial_{y}v - \partial_{y}u\,\partial_{x}v) / \pi,\ Q = ½\,\Sigma\,\rho \tag{8}$$

$$\mathcal{L}_{topo} = E_{GL} + w_{q}\,(Q - q^{*})^{2} \tag{9}$$

E_{GL}	Ginzburg–Landau energy of the field
$\langle \cdot \rangle$	Average over all pixels
ε	Typical size of a defect core in pixels (1.5)
ρ	Soft charge density, from the Jacobian of (u, v)
Q	Net index of the whole field (smooth estimate)
q^*	Target net index: 0 for arches, loops and whorls
w_q	Weight of the charge term (0.1)

In plain words: the first part of E_{GL} penalises rough changes in the field. The second part wants $|\Psi| = 1$ everywhere, but at the centre of every defect $|\Psi|$ must drop to 0, so **every defect costs energy**. Spurious defects therefore raise the loss, and the network learns to remove them. The charge density ρ is a smooth version of the index counter: it concentrates around vortices and integrates to their total charge. Because both parts are smooth, gradient descent can use them.

Why $q^* = 0$? An arch has no defects. A loop has one core (+½) and one delta (−½). A whorl has two of each. In every case the net index is 0, which matches the biology of a finger: the total turning of the ridges is fixed by the shape of the fingertip.

KEY IDEA

From zero gradient to a working loss

The original design counted defects directly and compared the count with a target. Counting produces whole numbers, so its gradient is zero almost everywhere: the loss could see mistakes but could not fix them. Measured on a test field, the old loss had a gradient norm of 0.00; the new one has 0.44. On a noisy loop with 37 core–delta pairs, the new loss removed the 36 spurious pairs and kept the true one.

4.8 The complete training objective

$$\mathcal{L} = MSE(\Psi, \Psi^*) + 0.05 \cdot \mathcal{L}_{topo}(\Psi) \tag{10}$$

In plain words: the network is trained to match the true field (the mean-squared-error term) while keeping the topology clean (the topological term). The weight 0.05 keeps the topology term in a supporting role: it shapes the answer without overpowering the data.

CHAPTER 05 · PART A · THEORY

5

Architecture and algorithm

Four stages, sixty wave steps and only 29 extra parameters.



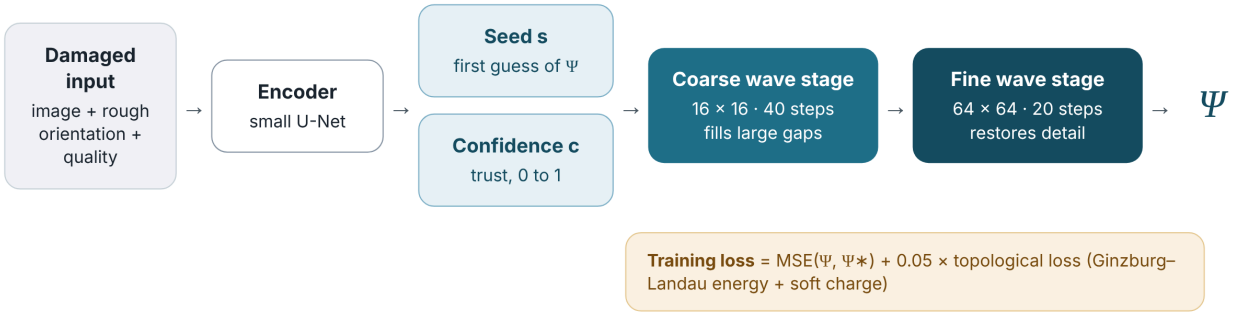


FIGURE 4 · The FingerWave pipeline. The encoder predicts a seed field and a confidence map. A coarse PDE stage fills large gaps; a fine stage restores local detail. Training combines a data term with the topological loss.

5.1 The four stages

1. **Encoder.** A small three-level U-Net reads the input (by default four channels: the image, the two channels u and v of a rough orientation estimate, and its coherence) and outputs three channels: a seed field s (two channels) and a confidence map c (one channel, squashed to 0–1 by a sigmoid).
2. **Coarse wave stage.** Seed and confidence are averaged down by 4× (a 64 × 64 field becomes 16 × 16). The PDE runs for 40 steps, starting from the seed. At low resolution, waves cross the whole field quickly, so this stage fills large holes.
3. **Blend.** The coarse result is upsampled back to full size and mixed with the seed: $\Psi = c \cdot s + (1 - c) \cdot \Psi_{coarse}$. Trusted pixels keep the encoder's estimate; untrusted pixels take the wave solution.
4. **Fine wave stage.** A second PDE runs for 20 steps at full resolution with seeding, restoring sharp local detail. The output is the repaired field Ψ.

5.2 The algorithm in pseudocode

```
function FingerWave(x):
    s, c_raw = Encoder(x)              # seed (2 ch), confidence (1 ch)
    c   = sigmoid(c_raw)
    λ   = 4 · sigmoid(λ_raw)           # one λ per stage, λ·Δt ≤ 0.8

    # coarse stage: long-range fill
    s4, c4 = avgpool(s, 4), avgpool(c, 4)
    Ψ = s4
    repeat 40 times:
        Ψ = PDE_step(Ψ)                # Eq. 6a + 6b
        Ψ = Ψ + Δt·λ₁·c4·(s4 - Ψ)      # Eq. 6c, boundary seeding

    # blend and fine stage: local detail
    Ψ = c·s + (1 - c)·upsample(Ψ, 4)
    repeat 20 times:
        Ψ = PDE_step(Ψ)
        Ψ = Ψ + Δt·λ₂·c·(s - Ψ)
    return Ψ                           # θ = ½·atan2(Ψ_v, Ψ_u)
```

5.3 What is learned

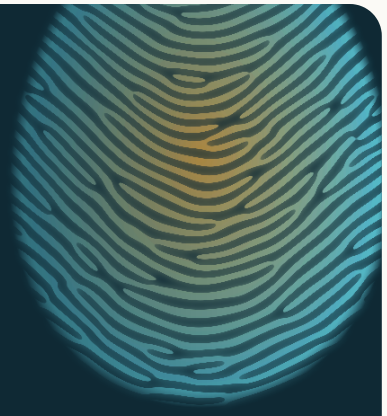
COMPONENT	LEARNED PARAMETERS	TYPICAL LEARNED VALUE (SEED 0)
Encoder (U-Net)	118,451 convolution weights	-
Coarse PDE	$d_{\parallel}$, $d_{\perp}$, α, β, γ	$d_{\parallel}$ 0.78, $d_{\perp}$ 0.32
Fine PDE	$d_{\parallel}$, $d_{\perp}$, α, β, γ	$d_{\parallel}$ 0.93, $d_{\perp}$ 0.33
Seeding	λ_1, λ_2	2.12, 1.80

The whole physics part of FingerWave adds only **12 parameters**, and the third encoder output channel adds 17 more. That is why FingerWave has just 29 more parameters than the baseline U-Net. The improvement comes from **structure**, not from size.

CHAPTER 06 · PART A · THEORY

6 How FingerWave compares with other AI models

What other model families assume, and what FingerWave adds.



FingerWave belongs to the family of physics-informed and PDE-based neural networks. The table below shows what each popular model family assumes about the data, and where FingerWave differs.

MODEL FAMILY	HOW IT SPREADS INFORMATION	RESPECTS TOPOLOGY?	USES TRUST / CONFIDENCE?
CNN	Fixed local filters, stacked in layers	No	No
U-Net	CNN with a coarse-to-fine path and skip connections	No	No
Vision Transformer	Attention between all image patches	No	Implicitly, through attention
Graph neural network	Messages between connected nodes	No	Sometimes, through edge weights
PINN	No spreading; a physics equation is added to the loss	Only if the physics says so	No
Neural operator (FNO)	Global spectral filters learned from data	No	No
Diffusion model (generative)	Iterative denoising from random noise	No	No
FingerWave	**Learned reaction–diffusion along the structure**	**Yes, through a differentiable defect loss**	**Yes, explicit confidence seeding**

What is genuinely new

Neural PDE layers (PDE-Net), reaction–diffusion graph networks (GREAD), neural cellular automata and topology-aware losses already exist individually. FingerWave's contribution is the **specific combination**, motivated by fingerprint development: confidence-seeded propagation, a stable complex Ginzburg–Landau wavefront layer that acts on a headless director field, and a differentiable defect-conserving loss, together with a benchmark that measures topology exactly.

FOR BUSINESS

How to say it in an interview

“Most models learn how to fill gaps from examples. FingerWave is told how gaps should be filled, by a physics layer that starts from trusted regions and conserves topology. The network only has to learn where to trust and how fast to spread.”

CHAPTER 07 · PART A · THEORY

7

Evidence: benchmark and results

500 damaged fingerprints, three seeds and one consistent winner.



7.1 How the benchmark was built

Measuring topology requires exact knowledge of where every core and delta is, and real fingerprint databases do not provide that. The benchmark therefore uses a synthetic generator in the style of SFinGe, the standard tool for synthetic fingerprints:

- **Orientation fields** from the Sherlock–Monro model, with randomly placed cores and deltas for five classes (arch, tented arch, left loop, right loop, whorl), random rotation and a smooth perturbation that does not change topology.
- **Ridges** rendered at 128 × 128 pixels by orientation-adaptive Gabor filtering.
- **Damage**: contrast loss, blur, strong noise, one to three smudges and a 50% chance of a missing band at one edge. On average, 14.9% of each image is hidden.
- **Split**: 2,000 training and 500 test prints. Every model receives the same inputs on a 64 × 64 grid.

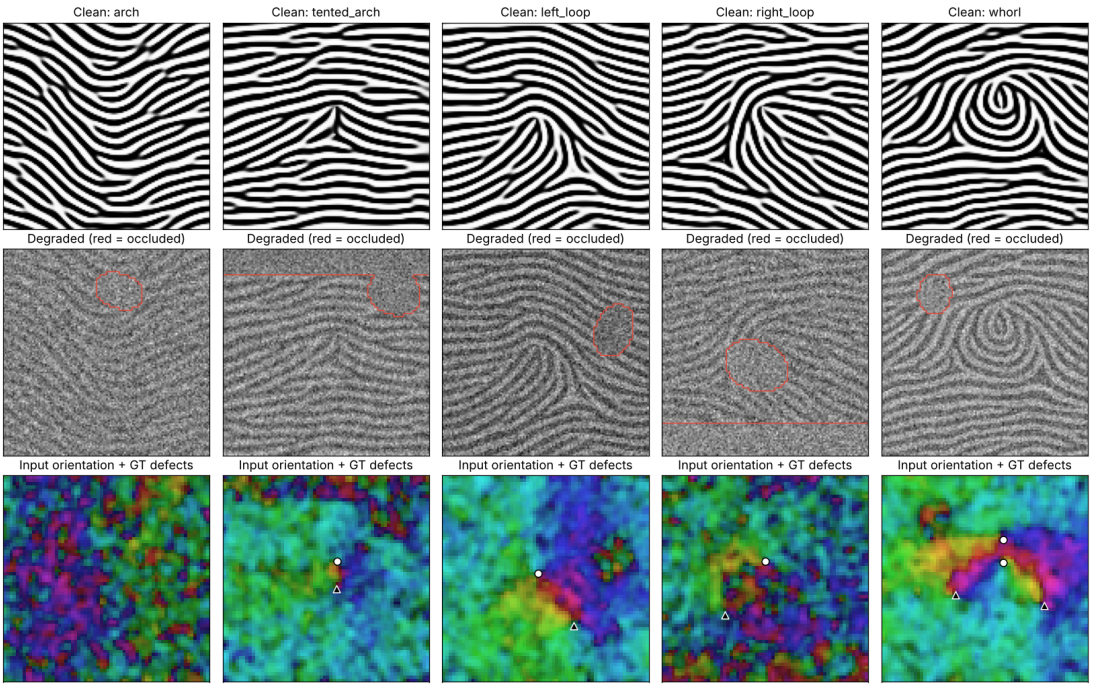


FIGURE 5 · One sample per class. Top: clean ridges. Middle: damaged input. Bottom: the noisy orientation estimate that every model receives, with true cores (○) and deltas (△).

7.2 Results

METHOD	ANGLE ERROR	HIDDEN REGIONS	TOPOLOGICAL ERROR RATE ↓	CLASS ACCURACY ↑
Input (no processing)	18.15°	41.12°	100%	0.0%
Classic smoothing	8.32°	16.98°	44.4%	55.6%
U-Net	4.79°	7.66°	5.8 ± 1.1%	94.2%
U-Net + topological loss	4.75°	7.62°	5.3 ± 0.5%	94.7%
FingerWave	**4.69°**	**7.36°**	**4.0 ± 0.2%**	**96.0%**

Values are means over three random seeds (± standard deviation). All learned models have about 118,000 parameters and were trained with identical settings. FingerWave's three runs scored 3.8%, 4.0% and 4.2%; the U-Net's scored 5.4%, 5.0% and 7.0%. Every FingerWave run beat every U-Net run.



FIGURE 6 · Angle error in hidden regions, Topological Error Rate and class accuracy (mean ± s.d., three seeds). Dark teal: FingerWave. Light teal: ablations. Grey: baselines.

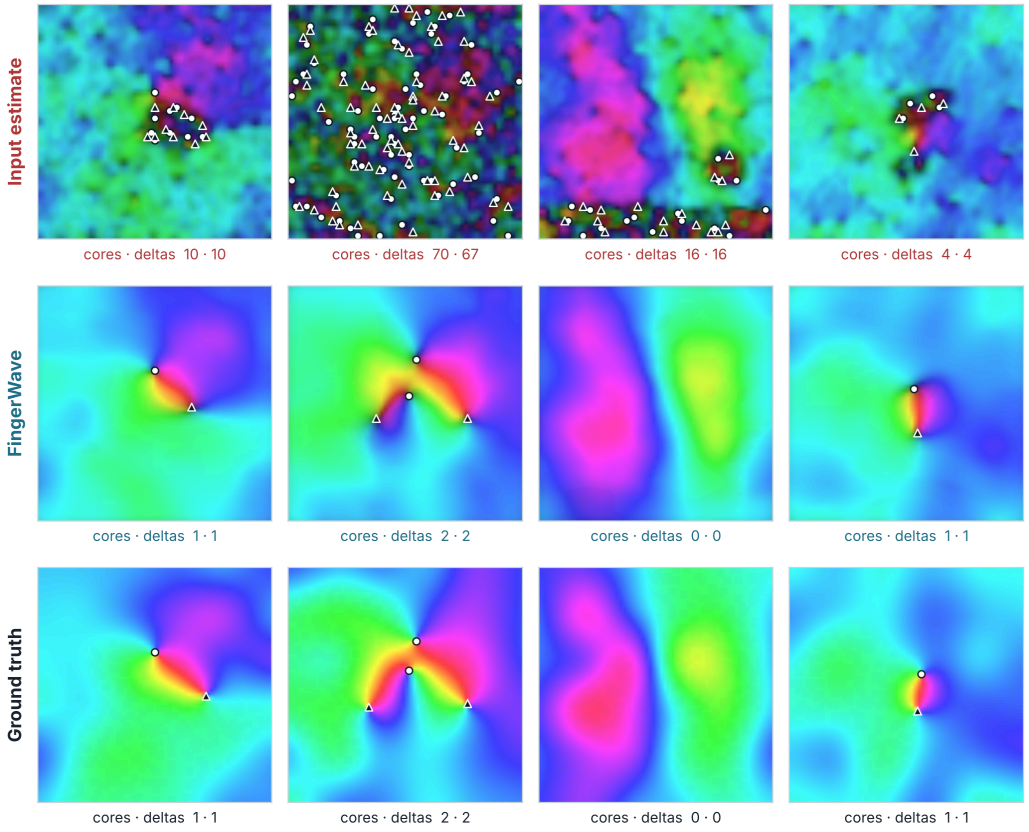


FIGURE 7 · The released package on four unseen test prints. The raw estimate (top) contains dozens of false cores and deltas; FingerWave (middle) recovers exactly the true counts (bottom). Colour shows ridge direction.

> *Every FingerWave run beat every U-Net run.*

7.3 What the ablations teach

COMPONENT REMOVED	WHAT HAPPENED	CONCLUSION
Boundary seeding	Error rate rose from 4.0% to 9.8%, 82.2% and 21.0% across seeds; one run collapsed	**Essential.** Starting from trust is the core idea

COMPONENT REMOVED	WHAT HAPPENED	CONCLUSION
Anisotropic diffusion	Matched the full model (3.9% vs 4.0%)	No extra benefit on this benchmark; the encoder already supplies direction
Topological loss (on a U-Net)	Error rate 5.8% → 5.3%	Helpful, and strongest when combined with seeding

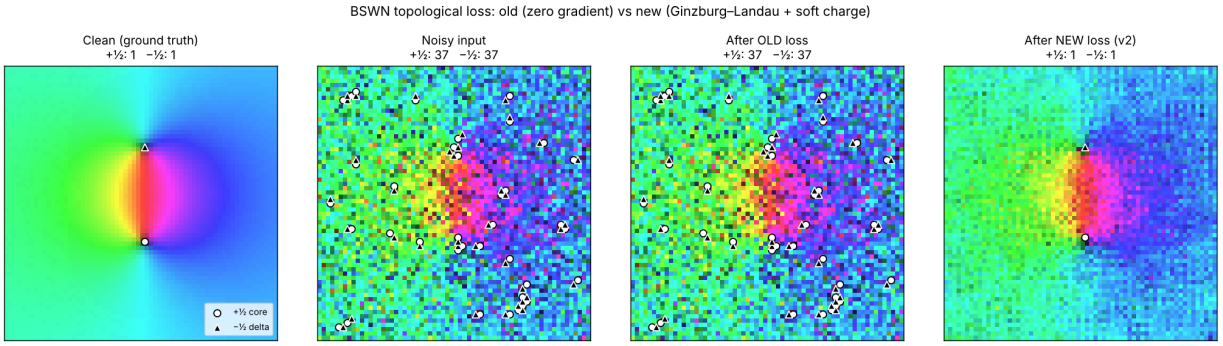


FIGURE 8 · The topological loss in isolation. Starting from a noisy loop with 37 core–delta pairs, the old counting loss changes nothing (zero gradient). The FingerWave loss removes the 36 spurious pairs and keeps the true core and delta.

7.4 Cost

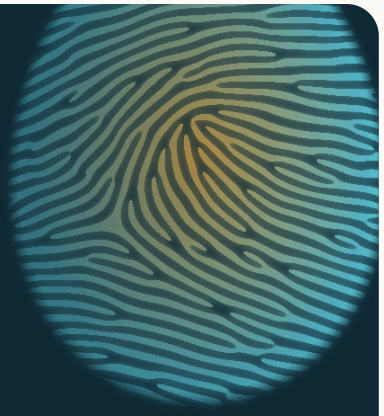
On a 2-core CPU, FingerWave trained in 449 seconds versus 125 seconds for the U-Net (about 3.6×) and ran in 4.1 ms per image versus 2.6 ms. The extra time comes from the 60 PDE steps. Both are small enough for real-time use on ordinary hardware, and faster PDE solvers are on the roadmap (Chapter 18).

CHAPTER 08 · PART A · THEORY

8

Using FingerWave in Python

From pip install to your own data in four steps.



8.1 Install and run

```
pip install "pavan-krovvidi-fingerwave[viz]"

import torch
import pavan_krovvidi_fingerwave as pk
from pavan_krovvidi_fingerwave import PavanKrovvidiFingerWaveNetwork
from pavan_krovvidi_fingerwave.data import make_dataset, make_model_input
from pavan_krovvidi_fingerwave.metrics import evaluate

pk.about()                                  # author, DOI, components
data  = make_dataset(8, seed=123)           # 8 new test fingerprints
x     = make_model_input(data)              # (8, 4, 64, 64)
model = PavanKrovvidiFingerWaveNetwork.from_pretrained()
with torch.no_grad():
    field = model(x)                        # (8, 2, 64, 64)
print(evaluate(field, data))                # angle error, precision, recall, TER
```

The same code runs unchanged in Google Colab. The pretrained weights ship inside the package, so no download is needed.

8.2 What is in the package

MODULE	PROVIDES
`network`	PavanKrovvidiFingerWaveNetwork (with from_pretrained), UNet baseline
`pde`	WavefrontPDELayer, diffusion tensor, flux-form divergence
`topo_loss`	topological_loss, ginzburg_landau_energy, soft_charge_density, count_defects
`data`	Synthetic fingerprint generator with exact ground truth
`metrics`	Angle error, core/delta precision and recall, TER, class accuracy
`viz`	orientation_image, plot_field

8.3 Adapting FingerWave to your own data

1. **Decide what Ψ is.** For headless orientations (fibres, ridges, layers), use the doubled angle $\Psi = A \cdot e^{i2\theta}$. For true directions with a head (wind, flow, movement), use the single angle $\Psi = A \cdot e^{i\theta}$; defects then become ±1 vortices. For a plain quantity (temperature, potential), use one channel and keep only the diffusion part.
2. **Prepare the inputs.** Give the encoder whatever helps it judge quality: the raw signal, a rough estimate of the field and a quality measure (coherence, signal-to-noise ratio, sensor presence). Set `in_channels` to match.
3. **Choose the constraint.** Set the target net index q^* for your domain, or replace the charge term with the rule your domain must obey.
4. **Train and evaluate structurally.** Report pixel error and a structural metric (such as the number of defects, vortices or hotspots that are wrong), and compare against a same-size U-Net.

```
from pavan_krovvidi_fingerwave import PavanKrovvidiFingerWaveNetwork, topological_loss

model = PavanKrovvidiFingerWaveNetwork(in_channels=5)    # your inputs
opt   = torch.optim.Adam(model.parameters(), lr=2e-3)

for x, target in loader:
    pred = model(x)
    loss = ((pred - target) ** 2).mean() + 0.05 * topological_loss(pred, target_index=0.0)
    opt.zero_grad(); loss.backward(); opt.step()
```

PART B

B Industry

Where FingerWave fits, the business problems it can address in seven industries, and how to run a first project.



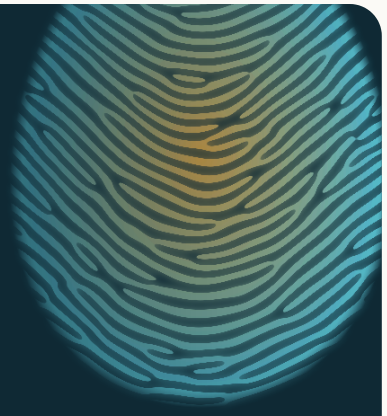
9	The FingerWave fit test	25
10	Biometrics and forensics: the home domain	26
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CHAPTER 09 · PART B · INDUSTRY

9

The FingerWave fit test

Three questions that tell you whether FingerWave fits your data.



FingerWave is a specialist, not a general-purpose model. It is at its best when three conditions hold together. Before starting a project, ask these three questions about your data.

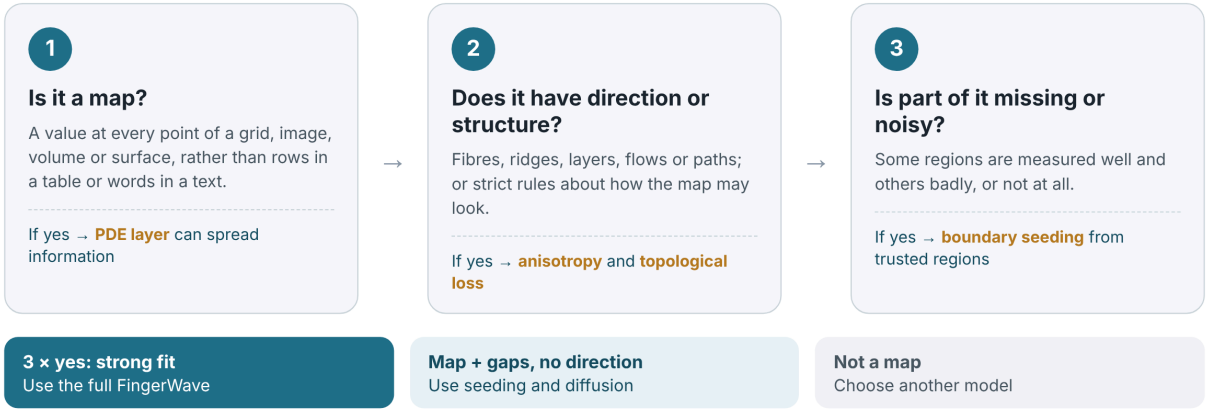


FIGURE 9 · The FingerWave fit test. Three yes answers indicate a strong fit; the right-hand column shows which part of FingerWave each answer activates.

DATA TYPE	MAP?	DIRECTION OR STRUCTURE?	GAPS OR NOISE?	FIT
Fingerprints, brain fibres, seismic layers	Yes	Yes (orientation)	Yes	**Strong**
Airflow, customer flow, ocean currents	Yes	Yes (direction)	Yes	**Strong** (single-angle version)
Temperature, electrostatic potential	Yes	Structure only	Yes	**Good** (diffusion and seeding)
Volatility surfaces, risk maps	Yes (abstract axes)	Constraints only	Yes	**Good** (seeding and constraint loss)
Tables, text, clinical codes, transactions	No	-	-	Use another model

GOOD TO KNOW

The three parts are modular

Each FingerWave component can be used alone. **Seeding** helps whenever some measurements are more reliable than others. The **PDE layer** helps whenever information should spread in a physically sensible way. The **topological loss** helps whenever the output must contain the right number of special points.

CHAPTER 10 · PART B · INDUSTRY

10

Biometrics and forensics: the home domain



Crime scenes, national databases and the phone in your pocket.

Fingerprints are FingerWave's home ground. Fingerprint identification is used by police forces, border control, national ID programmes, banks and every smartphone with a fingerprint sensor. The global biometrics market runs into tens of billions of dollars, and fingerprints remain its most widely deployed modality.

Latent prints at crime scenes

PROBLEM	Prints lifted from crime scenes are partial, smudged and overlapped with background patterns. Examiners spend hours marking ridge flow by hand, and automatic systems often fail on the worst prints, which are exactly the ones that matter in difficult cases.
HOW FINGERWAVE HELPS	FingerWave reconstructs the ridge orientation field from the reliable parts of the print and propagates it into smudged or missing regions while keeping the correct number of cores and deltas. A clean orientation field is the foundation for ridge enhancement, minutiae extraction and matching.
BUSINESS VALUE	Fewer prints rejected as unusable, less manual marking time per case, and fewer false leads caused by spurious singular points.

Faster search in large databases

PROBLEM	National databases hold hundreds of millions of records. Searches are narrowed by pattern class (arch, loop, whorl) and by the positions of cores and deltas. Misclassification sends the search to the wrong part of the database.
HOW FINGERWAVE HELPS	FingerWave recovered the correct class for 96% of damaged benchmark prints, directly from its repaired topology, compared with 94.2% for a same-size U-Net and 55.6% for classic smoothing.
BUSINESS VALUE	Smaller candidate lists, faster searches and lower compute cost per query.

Phone and border sensors

PROBLEM	Small sensors capture only part of the finger, and wet, dry or worn fingers produce broken ridges. Users then see repeated rejections at a phone unlock or a border gate.
HOW FINGERWAVE HELPS	The seeding mechanism is designed for exactly this case: it trusts the clean patch and extends the pattern outward along the ridge flow. The model is small (about 118,000 parameters) and runs in milliseconds on a CPU.
BUSINESS VALUE	Fewer false rejections, a smoother user experience and fewer manual checks at gates.

FINGERWAVE MAPPING

APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Biometrics	Ridge orientation (doubled angle)	Ridge clarity, sensor contact	Cores and deltas	Net index 0

CHAPTER 11 · PART B · INDUSTRY

11

Healthcare and medical imaging

Brain fibres, retinal vessels, heart muscle and tumour collagen.



Many medical images are **orientation fields** in disguise. Nerve fibres in the brain, muscle fibres in the heart, blood vessels in the retina and collagen fibres in tumour tissue all have a local direction without a head or tail, exactly like fingerprint ridges. They are also often noisy or incomplete because of short scan times, patient motion and imaging artefacts.

Brain fibre maps from diffusion MRI

PROBLEM	Diffusion MRI measures how water moves through brain tissue, which reveals the direction of nerve fibre bundles. Neurosurgeons use the resulting fibre maps (tractography) to plan tumour removal without damaging critical pathways such as language and motor tracts. High-quality maps need long scans with many measurement directions, and motion or noise corrupts regions of the map.
HOW FINGERWAVE HELPS	Fibre direction is a director field, so FingerWave's doubled-angle representation fits it directly. The confidence map can be guided by the signal-to-noise ratio and fractional anisotropy (how strongly directional each voxel is). Seeding anchors the well-measured white matter, and the anisotropic PDE propagates fibre direction along the bundles into corrupted voxels.
BUSINESS VALUE	Shorter scans for the same map quality, fewer repeat scans after motion, and more reliable maps for surgical planning.

Retinal vessel analysis

PROBLEM	Retinal photographs are used to screen millions of people for diabetic retinopathy, and the shape and twistiness of retinal vessels are markers of diabetes and high blood pressure. Lesions, reflections and poor focus interrupt vessels, which breaks automatic vessel tracing.
HOW FINGERWAVE HELPS	The vessel orientation field is reconstructed from clear vessel segments and propagated across interruptions along the vessel direction. The topological loss discourages broken or invented junctions.
BUSINESS VALUE	More complete vessel maps, more reliable automatic measurements and fewer images sent for manual review.

Heart muscle fibre orientation

PROBLEM	Patient-specific computer models of the heart are used to plan treatments for arrhythmias such as atrial fibrillation. These models need the direction of muscle fibres at every point, which is very hard to measure in a living patient, so rule-based estimates are often used instead.
HOW FINGERWAVE HELPS	Sparse or noisy fibre measurements act as seeds, and the PDE layer completes a smooth, physically consistent fibre field between them.
BUSINESS VALUE	More personalised heart models, which can support better ablation planning.

Collagen alignment in cancer pathology

PROBLEM	In breast cancer, the alignment of collagen fibres around a tumour has been linked to patient outcome: fibres running straight out from the tumour are associated with invasion. Measuring alignment by eye is slow and subjective.
HOW FINGERWAVE HELPS	FingerWave extracts a clean collagen orientation field from microscopy images, even where staining is uneven, and quantifies alignment and singular points consistently.
BUSINESS VALUE	A reproducible, quantitative biomarker that can support pathologists' decisions.

> *A model that shows what it trusted is a model a clinician can question.*

FINGERWAVE MAPPING

APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Brain MRI	Fibre direction (doubled angle)	Signal-to-noise ratio, anisotropy	Fibre crossings and fanning	Smoothness along tracts

APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Retina	Vessel direction	Vessel contrast	Junctions	Connected vessel tree
Heart	Muscle fibre direction	Measurement density	Rare	Smooth helical structure

FOR BUSINESS

Why this matters for healthcare AI teams

Regulators and clinicians are wary of AI that hallucinates anatomy. A model that starts from measured data, shows which regions it trusted (the confidence map) and is mathematically prevented from inventing structure is easier to validate and to explain.

CHAPTER 12 · PART B · INDUSTRY

12

Drug discovery and life sciences

Electrostatics, cryo-EM maps and the shape of living tissue.



Drug discovery is full of **fields**: the electric potential around a protein, the electron density in a cryo-EM map, the arrangement of cells in a tissue. Computing or measuring these fields is slow and expensive, and many are incomplete. FingerWave's physics layer is closely related to the equations that describe them.

Fast electrostatics for virtual screening

PROBLEM	Electrostatic interactions are a major part of how a drug binds to its target. They are computed with the Poisson–Boltzmann equation, using solvers such as APBS. Each calculation can take seconds to minutes, which is too slow when screening millions of candidate molecules.
HOW FINGERWAVE HELPS	The core of the Poisson–Boltzmann equation, $\nabla\cdot(\varepsilon\nabla\varphi)$, has exactly the same form as FingerWave's diffusion term $\nabla\cdot(D\nabla\Psi)$. The fixed charges on the atoms play the role of seeds, and the dielectric map ε plays the role of the diffusion tensor. A FingerWave-style network can learn a fast surrogate that keeps the physical structure of the solution.
BUSINESS VALUE	Electrostatic scoring at a fraction of the cost, so more molecules can be screened with physics-aware accuracy.

Completing cryo-EM density maps

PROBLEM	Cryo-electron microscopy produces 3D density maps of proteins, but resolution varies across the map: the rigid core is sharp while flexible loops are blurred. Model building in the blurred regions is slow and error-prone.
HOW FINGERWAVE HELPS	Local-resolution maps are a natural confidence map. Seeding anchors the high-resolution core, and the PDE propagates structure outward along helices and strands, which have a clear direction.
BUSINESS VALUE	Faster model building, more complete structures and quicker cycles in structure-based drug design.

Cell organisation and drug phenotypes

PROBLEM	Cells grown in a dish often align like liquid crystals, and their ±½ topological defects have been linked to where cells are squeezed out of the layer. Changes in cell alignment are also a readout in drug screening.
HOW FINGERWAVE HELPS	FingerWave's defect-aware loss is built for exactly these ±½ defects. It can produce clean orientation fields and reliable defect counts from noisy microscopy, frame after frame.
BUSINESS VALUE	A new, quantitative readout for high-content screening and tissue engineering.

Protein surface properties

PROBLEM	Binding sites are judged by properties on the curved surface of a protein, such as charge and water-repellence, which are often noisy or incomplete.
HOW FINGERWAVE HELPS	Extending the PDE layer from a flat grid to a curved surface (with the Laplace–Beltrami operator) lets the same seed-and-propagate idea run directly on molecular surfaces.
BUSINESS VALUE	Cleaner surface maps for binding-site prediction.

FINGERWAVE MAPPING

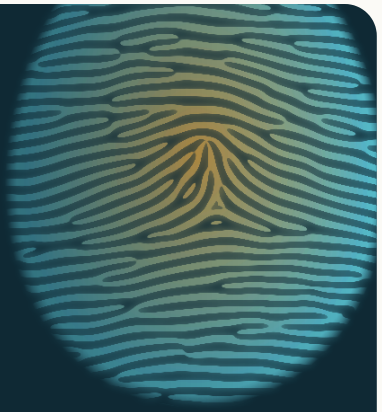
APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Electrostatics	Potential φ (one channel)	Known charges	-	Physics of the solver
Cryo-EM	Density and helix direction	Local resolution	Chain ends, branch points	Chain connectivity
Cell layers	Cell orientation (doubled angle)	Image quality	±½ defects	Net index

CHAPTER 13 · PART B · INDUSTRY

13

Automobile and mobility

Airflow, battery heat, lanes and carbon-fibre parts.



Modern vehicles are designed with simulations and run on sensors. Both produce fields: airflow around the body, temperature inside a battery pack, the direction of lanes on the road. Simulations are slow and sensors are sparse, which is exactly the setting FingerWave was built for.

Fast aerodynamic previews

PROBLEM	Computational fluid dynamics (CFD) simulations of airflow around a car can take hours each. Designers want instant feedback when they change a shape, so teams train AI surrogates, but plain networks can produce flow fields with vortices in the wrong places.
HOW FINGERWAVE HELPS	Flow direction is a true direction, so FingerWave's single-angle version ($\Psi = A \cdot e^{i\theta}$) applies, with vortices as ±1 defects. Coarse simulations or a few probe measurements act as seeds, and the defect-aware loss keeps the number and position of vortices right.
BUSINESS VALUE	Interactive design iteration, fewer full simulations and earlier detection of drag and noise problems.

Battery thermal maps

PROBLEM	An electric vehicle battery pack has many cells but only a limited number of temperature sensors. Hotspots between sensors can go unnoticed, and hotspots are an early warning for degradation and thermal runaway.
HOW FINGERWAVE HELPS	Heat flow is a diffusion process, so FingerWave's diffusion layer matches the physics. Each sensor is a high-confidence seed, and the anisotropic diffusion tensor can follow cooling channels and cell layout.
BUSINESS VALUE	A full temperature map from existing sensors, earlier hotspot warnings and better battery-management decisions.

Lane direction under occlusion

PROBLEM	Driver-assistance and self-driving systems need the direction of every lane, but lane markings are hidden by other vehicles, worn away, or covered by shadows and rain.
HOW FINGERWAVE HELPS	Visible lane segments are seeds. The orientation field is propagated along the road into hidden regions, and junctions and merges behave like singular points.
BUSINESS VALUE	More stable lane estimates in traffic and bad weather.

Fibre orientation in composite parts

PROBLEM	Carbon-fibre and glass-fibre parts are strong only if the fibres run in the intended directions. Inspection images are noisy, and wrinkles or misaligned regions are defects that must be found.
HOW FINGERWAVE HELPS	Fibre direction is a director field. FingerWave produces a clean orientation map and highlights singular points that indicate wrinkles or misalignment.
BUSINESS VALUE	Faster quality inspection and fewer faulty parts reaching assembly.

FINGERWAVE MAPPING

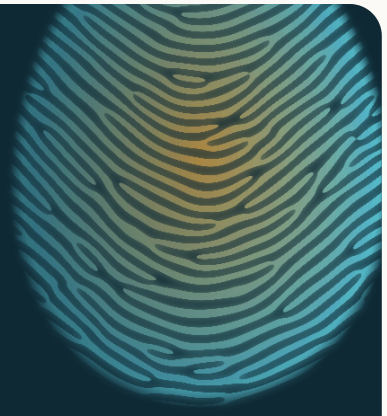
APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Airflow	Flow direction (single angle)	Probe and coarse-simulation data	Vortices	Flow physics
Battery	Temperature (one channel)	Sensor locations	Hotspots	Heat diffusion
Lanes	Lane direction (doubled angle)	Marking visibility	Merges, junctions	Road geometry

CHAPTER 14 · PART B · INDUSTRY

14

Finance and insurance

Volatility surfaces and risk maps with gaps.



Finance may look far from fingerprints, but several of its core objects are **surfaces with gaps** that must obey strict rules. In these applications the axes of the map are abstract (strike price and maturity, or latitude and longitude), and the parts of FingerWave that carry over are confidence seeding, smooth propagation and a constraint loss.

Completing volatility surfaces

PROBLEM	Option prices define an implied-volatility surface over strike price and time to maturity. Quotes are dense for popular contracts and sparse or stale for others, yet traders and risk systems need a complete, smooth surface. The surface must also be free of arbitrage: certain shapes are impossible because they would allow risk-free profit.
HOW FINGERWAVE HELPS	Liquid, recent quotes are high-confidence seeds (confidence can come from bid–ask spread and trading volume). The PDE layer fills the sparse regions smoothly, and the constraint loss, which plays the role of the topological loss, penalises shapes that violate no-arbitrage rules.
BUSINESS VALUE	More reliable pricing and risk numbers for illiquid contracts, with fewer manual fixes to the surface.

Spatial risk maps for insurance and lending

PROBLEM	Insurers and lenders price flood, wildfire and heat risk by location. Observations such as sensors, claims and inspections are scattered, and water and fire spread along the terrain rather than equally in every direction.
HOW FINGERWAVE HELPS	Observations are seeds, and the anisotropic diffusion tensor follows terrain slopes or river directions, so risk spreads along the paths that water or fire would actually take.
BUSINESS VALUE	Finer, more defensible risk pricing and better portfolio stress tests.

Property value maps

PROBLEM	Property values are known only where sales happened recently. Valuers and lenders need an estimate for every location.
HOW FINGERWAVE HELPS	Recent sales are seeds weighted by recency, and diffusion follows roads and neighbourhood boundaries rather than straight lines.
BUSINESS VALUE	More consistent automated valuations.

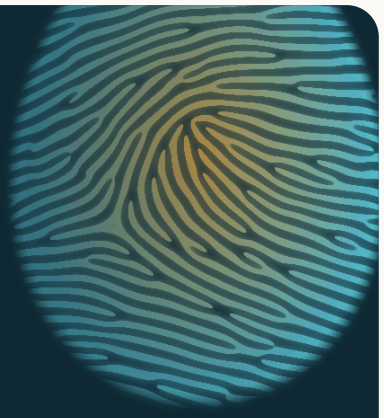
FINGERWAVE MAPPING

APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Volatility	Implied volatility (one channel)	Liquidity, quote age	-	No-arbitrage shape rules
Risk maps	Risk score (one channel)	Observation density	Hotspots	Terrain-following spread

CHAPTER 15 · PART B · INDUSTRY

15 E-commerce, retail and logistics

Customers, robots and riders: fields of movement.



Physical retail and logistics are about **movement**: shoppers walking through a store, robots driving through a warehouse, riders moving through a city. Movement fields are measured only partially, by cameras with blind spots or by scattered GPS points, and the important features are where flows split, merge or get stuck.

Customer flow in stores and malls

PROBLEM	Retailers place products, staff and signs based on how customers move. Cameras cover only part of the floor, and blind spots hide exactly where queues and dead zones form.
HOW FINGERWAVE HELPS	Customer movement is a direction field (single angle). Tracked camera regions are seeds, and the PDE completes the flow map across blind spots. Points where flow circulates or stalls appear as defects, which highlights congestion and dead zones.
BUSINESS VALUE	Better layout and product placement, shorter queues and staffing matched to real flow.

Warehouse robot traffic

PROBLEM	Fulfilment centres run fleets of mobile robots on shared floor space. Traffic rules are often expressed as direction fields (which way robots may travel on each tile), and conflicts create deadlocks.
HOW FINGERWAVE HELPS	FingerWave produces smooth, consistent direction fields from partial rules or observed traffic, and the defect-aware loss discourages points where directions conflict.
BUSINESS VALUE	Fewer deadlocks, higher throughput and faster re-planning when the layout changes.

Delivery demand maps

PROBLEM	Quick-commerce companies decide where to place dark stores and riders from scattered historical orders. Demand spreads along roads, not in circles.
HOW FINGERWAVE HELPS	Order locations are seeds, and the anisotropic diffusion tensor follows the road network, which gives a realistic demand surface for every street.
BUSINESS VALUE	Better store placement, shorter delivery times and lower cost per order.

FINGERWAVE MAPPING

APPLICATION	FIELD Ψ	CONFIDENCE C (SEEDS)	DEFECTS	CONSTRAINT
Store flow	Movement direction (single angle)	Camera coverage	Circulation and stall points	Walls and aisles
Robots	Travel direction	Rule or observation density	Conflict points	Collision-free flow
Demand	Order density (one channel)	Order history	Demand peaks	Road network

CHAPTER 16 · PART B · INDUSTRY

16

Energy, earth and materials

Rock layers, satellite gaps, crop rows and crystals.



Seismic interpretation

PROBLEM	Oil and gas, geothermal and carbon-storage projects depend on seismic images of underground rock layers. Interpreters trace layer orientation and faults through noisy, partly missing data, which takes weeks for a large survey.
HOW FINGERWAVE HELPS	Rock layers have an orientation (dip) without a head or tail, which is a director field. High-coherence regions are seeds, and orientation is propagated along the layers. Faults and layer endings show up as structural defects.
BUSINESS VALUE	Faster interpretation and more reliable models of the subsurface.

Satellite gap filling

PROBLEM	Clouds hide large parts of satellite images of sea temperature, ocean colour and land surfaces. Ocean currents and fronts have clear directions.
HOW FINGERWAVE HELPS	Cloud-free pixels are seeds, and the PDE fills cloudy regions along currents and fronts instead of blurring across them.
BUSINESS VALUE	Complete daily maps for fisheries, climate monitoring and shipping.

Crop rows from drone images

PROBLEM	Farm robots and yield maps need the direction of crop rows, which is interrupted by missing plants, weeds and shadows.
HOW FINGERWAVE HELPS	Crop rows are an orientation field. Clear rows are seeds, and FingerWave continues the row direction through gaps.
BUSINESS VALUE	More reliable navigation for farm robots and accurate plant counts.

Materials and liquid crystals

PROBLEM	Metal grain orientation maps from electron microscopy contain unreadable pixels, and liquid-crystal displays are inspected for ±½ defects.
HOW FINGERWAVE HELPS	Both are orientation fields with natural ±½ defects, which is the case FingerWave's loss was designed for.
BUSINESS VALUE	Faster materials characterisation and automated defect inspection.

CHAPTER 17 · PART B · INDUSTRY

17

Business value and adoption playbook

Five value drivers, a six-week pilot and the KPIs that matter.



— 17.1 Five value drivers

VALUE DRIVER	WHAT IT MEANS FOR A BUSINESS
Structural correctness	Fewer impossible outputs, which means less manual review and fewer downstream errors
Works with incomplete data	Fewer rescans, re-measurements and rejected samples
Built-in explainability	The confidence map shows which inputs the model trusted, which helps audits and regulators
Small and fast	About 118,000 parameters and milliseconds per image on a CPU, so it can run on edge devices
Physics-aligned	Behaviour follows a known equation, which builds trust in regulated industries

— 17.2 A first project in six steps

1 **Frame the field** WEEK 1

Define Ψ (orientation, direction or quantity), the confidence signal and the structural rule your output must obey.

2 **Audit the data** WEEKS 1–2
Find ground truth: annotated cases, high-quality reference scans or simulations. If none exist, build a synthetic benchmark, as was done for fingerprints.

3 **Build a baseline** WEEK 2
Train a same-size U-Net so the comparison is fair.

4 **Adapt FingerWave** WEEKS 3–4
Set the input channels and the representation, and choose the constraint term.

5 **Measure structure** WEEK 5
Report pixel accuracy and a structural error rate, such as wrong defect counts, missing vortices or missed hotspots.

6 **Integrate** WEEK 6
Wrap the model behind an API and show the confidence map to users.

— 17.3 KPIs by industry

INDUSTRY	TECHNICAL KPI	BUSINESS KPI
Biometrics	Topological Error Rate, class accuracy	Rejected prints, search time per query
Healthcare	Fibre-angle error, tract completeness	Scan time, repeat-scan rate, review time
Drug discovery	Error against the physics solver	Molecules screened per day, cost per molecule
Automobile	Vortex count and position error, hotspot recall	Simulations saved, design cycle time
Finance	Arbitrage violations, fit to quotes	Manual surface fixes, pricing disputes
Retail and logistics	Flow-direction error, conflict points	Queue time, deadlocks, delivery time

FOR BUSINESS

The 30-second pitch

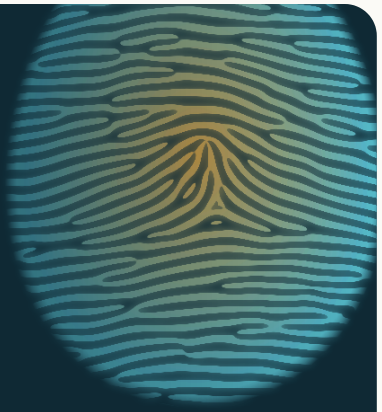
“Your data has gaps and your current AI fills them with structure that cannot exist. FingerWave fills gaps the way nature does: it starts from what it can trust, spreads information along the structure and never invents the special points that matter. On fingerprints it cut structural errors by 31% with the same model size.”

CHAPTER 18 · PART B · INDUSTRY

18

Roadmap

Where FingerWave goes next, and how to join in.



DIRECTION	WHAT IT ADDS
Real fingerprint databases	Validation on public databases such as FVC and NIST, starting from the released pretrained model
GPU-scale training	Larger models, larger images and more random seeds
Faster PDE	Fewer steps through implicit or multigrid solvers, to close the speed gap with the U-Net
3D version	Volumes for MRI, CT, cryo-EM and seismic data
Curved surfaces	A Laplace–Beltrami PDE layer for protein surfaces and organ shapes
Single-angle and scalar variants	Ready-made versions for flow fields and quantities such as temperature
Industry pilots	Electrostatics surrogate for drug discovery as the first pilot, then medical imaging

Collaboration is welcome. If you work with imaging or field data in any of the industries in Part B and would like to test FingerWave on it, get in touch using the details below.

Let's collaborate

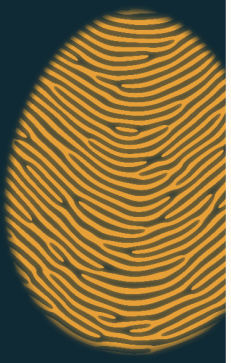
Questions, pilots, research collaboration or job opportunities: I'd be glad to hear from you.

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BACK MATTER

Appendices

Glossary, interview questions, notation, references and citation.



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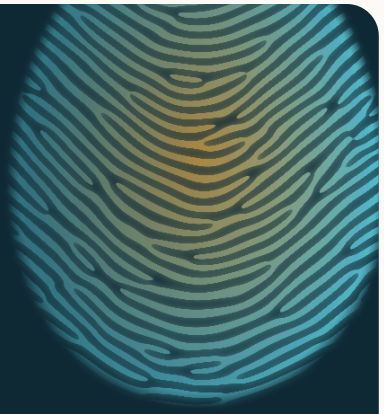
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APPENDIX A · APPENDICES

Glossary

The terms used in this guide, in one place.



TERM	MEANING
Ablation	Removing one part of a model to measure how much it contributes
Anisotropic	Behaving differently in different directions
Boundary seeding	Starting from trusted pixels and pulling them toward the encoder's estimate
Confidence map	A 0–1 score of how much each pixel is trusted
Core	A singular point where ridges turn back (index +½)
Delta	A singular point where three ridge systems meet (index −½)
Differentiable	Has a usable gradient, so it can be trained with gradient descent
Diffusion	Spreading of a quantity from high to low values
Director field	A field of headless directions (orientations)
Doubled angle	Storing 2θ instead of θ so that θ and θ+π become identical
Encoder	The first network stage that reads the input and extracts features
Field	A value attached to every point of a space
Ginzburg–Landau energy	An energy that makes every defect cost something
Gradient	The direction and rate in which a loss changes; used to train networks
Initiation site	A place where fingerprint ridges begin to form
Orientation	A direction without head or tail
PDE	Partial differential equation: a rule for how a field changes in space and time
Reaction–diffusion	Local creation plus spreading, which produces patterns
Semi-implicit	A time-stepping method that treats risky terms in a stable way
Singular point	A point where the orientation is undefined, such as a core or delta
Surrogate model	A fast AI model that imitates a slow simulation
Topological defect	A point with a non-zero index that cannot be removed by small changes

TERM	MEANING
Topological Error Rate (TER)	Share of images with the wrong number of cores or deltas
Turing pattern	A stripe or spot pattern produced by reaction–diffusion
U-Net	A popular encoder–decoder network for images

APPENDIX B · APPENDICES

B

Interview questions and answers

Twenty questions you may be asked, with answers.



Q1 **What is FingerWave in one sentence?**
A neural network that repairs damaged structural data the way fingertips grow ridges: it starts from trusted regions, spreads information along the structure with a learned PDE, and conserves topological defects with a differentiable loss.

Q2 **Why fingerprints?**
Fingerprint development is a natural example of building a complex, structurally correct pattern from a few reliable starting points. The same problem, completing structure from partial data without inventing features, appears in medicine, physics and engineering.

Q3 **What problem does it solve that a CNN does not?**
CNNs fill gaps locally and can create structures that are impossible as a whole, such as extra cores. FingerWave constrains how information spreads and penalises spurious defects, so its outputs stay structurally valid.

Q4 **What is a director field and why double the angle?**
A director field stores orientations without a head or tail. Doubling the angle maps θ and $\theta+\pi$ to the same value, which makes the field continuous and turns $\pm\frac{1}{2}$ defects into ordinary ±1 vortices.

Q5 **What is a topological defect?**
A point around which the orientation turns by a non-zero amount, such as a core ($+\frac{1}{2}$) or delta ($-\frac{1}{2}$). Its index cannot change by small perturbations; defects appear or vanish only in pairs.

Q6 **Why did the original topological loss have zero gradient?**
It counted defects, and a count is a step function: it is flat almost everywhere and jumps at a few points. Its gradient is zero almost everywhere, so training could not use it. I replaced it with Ginzburg–Landau energy and a soft Jacobian charge, which are smooth.

Q7 **How does seeding work?**
The encoder predicts a seed field and a confidence map. At every PDE step, each pixel is pulled toward the seed in proportion to its confidence. Trusted pixels act like initiation sites; untrusted pixels are filled by the waves.

Q8 **Why put a PDE inside a neural network?**
It gives the network a built-in rule for how information may spread. The network then only needs to learn where to trust and how fast to spread, which needs fewer parameters and generalises in a physically sensible way.

Q9 **How do you keep the PDE stable?**
Diffusion speeds are bounded with a sigmoid, the time step is 0.2/d_max (below the ¼ stability limit), the cubic reaction is treated semi-implicitly, and the seeding strength satisfies $\lambda \cdot \Delta t \leq 0.8$.

Q10 **What does the Ginzburg–Landau energy do?**
It rewards smooth fields with unit amplitude. Since the amplitude must fall to zero at a defect's centre, every defect costs energy, so spurious defects are pushed out during training.

Q11 **What were the results?**
On 500 degraded benchmark fingerprints, the Topological Error Rate fell from 5.8% (same-size U-Net) to 4.0%, class accuracy rose from 94.2% to 96.0%, and every FingerWave run beat every U-Net run across three seeds.

Q12 **Why use a synthetic benchmark?**
Topology must be measured exactly, and real databases do not record the exact positions of cores and deltas. A synthetic generator gives exact ground truth. Validation on real databases such as FVC and NIST is the next step.

Q13 **What did the ablations show?**
Seeding is essential: without it, the error rate rose to 37.7% on average and one run collapsed. Learned anisotropy added no measurable benefit on this benchmark.

Q14 **Why might anisotropy not have helped?**
The encoder already learns direction-aware features, so the PDE mainly needs to transport information rather than steer it. Anisotropy may matter more for larger gaps or data where direction dominates, such as diffusion MRI.

Q15 **How is FingerWave different from a PINN or a neural operator?**
A PINN adds a known equation to the loss; a neural operator learns a solution map from data. FingerWave puts a learnable PDE inside the network as a layer, adds confidence seeding and constrains topology, which neither of those does by default.

Q16 **What does it cost to run?**
About 4.1 ms per image versus 2.6 ms for the U-Net on a 2-core CPU, and about 3.6 times longer to train. Faster PDE solvers are on the roadmap.

Q17 **How would you apply it to drug discovery?**
As a fast surrogate for Poisson–Boltzmann electrostatics: the diffusion term matches $\nabla\cdot(\varepsilon\nabla\varphi)$, atom charges act as seeds and the dielectric map acts as the diffusion tensor. Cryo-EM map completion is a second option, with local resolution as the confidence map.

Q18 **How would you apply it to medical imaging?**
Fibre orientation in diffusion MRI is a director field, so the doubled-angle representation fits directly. Signal quality guides the confidence map, and the PDE propagates fibre direction along tracts into corrupted voxels.

Q19 **What would you do with more compute?**
Validate on real fingerprints, scale to GPUs and larger models, run more seeds, build the 3D version and test the anisotropy hypothesis on larger occlusions.

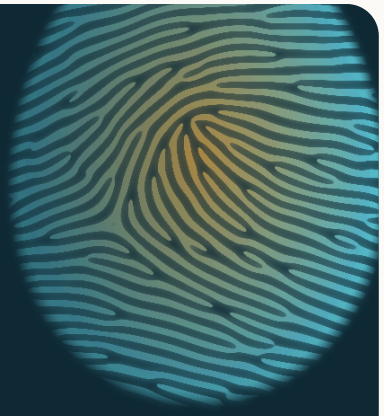
Q20 **What did you learn from building it?**
That evidence beats intuition. Three of my assumptions failed tests (the loss gradient, hard seeding and the stability of the reaction term), and fixing them made the method work. I used AI tools for coding and review; the idea, the design decisions and the research direction were mine.

APPENDIX C · APPENDICES

C

Notation

Every symbol and its default value.



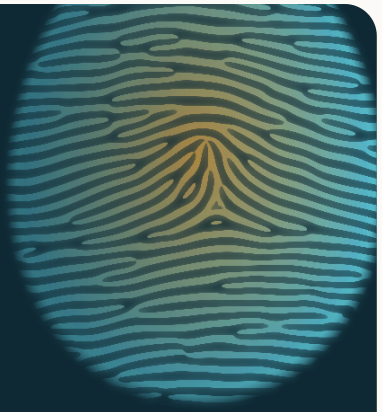
SYMBOL	MEANING	DEFAULT
θ	Ridge orientation	-
$\Psi = u + iv$	Doubled-angle field	-
A	Amplitude $\lvert\Psi\rvert$	-
c	Confidence map	0 to 1
s	Seed field	-
D	Diffusion tensor	-
$d_{\parallel}$, $d_{\perp}$	Diffusion along and across ridges	initial 0.9, 0.2
d_{max}	Bound on diffusion speed	1.0
Δt	Time step	0.2
α, β, γ	Ginzburg–Landau reaction coefficients	initial 1, 0, 0
λ	Seeding strength	≤ 4 ($\lambda \cdot \Delta t \leq 0.8$)
ε	Defect core size	1.5 pixels
w_q	Charge-term weight	0.1
q^*	Target net index	0
Steps	Coarse / fine PDE steps	40 / 20

APPENDIX D · APPENDICES

D

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APPENDIX E · APPENDICES

E

About the author, citation and license

Who built FingerWave, and how to cite it.





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AI assistance

I used AI assistance (Claude, Anthropic) during this work, for coding, testing, reviewing my design and writing. The ideas, decisions and research direction are my own. All results come from executed code and can be reproduced with the released package.



Every life has a unique print.
So does every prediction.

The Pavan Krovvidi FingerWave Neural Network is open source under the Apache License 2.0. Use it, build on it, and keep the name.

CODE ON GITHUB

CONNECT ON LINKEDIN

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pip install pavan-krovvidi-fingerwave · DOI 10.5281/zenodo.23766123 · ORCID 0009-0001-3286-7173